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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not detected.
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
Only mice that became pregnant three times were included in subsequent analyses.
Subject Demographics
Age: Exogenous P4 treatment study For P4 treatment studies, 12-week-old ovariectomized FVBn mice were purchased from the Jackson Laboratory (strain 001800).
Randomization
Briefly, mRNA was enriched using oligo(dT) beads, which was followed by two rounds of purification, and was fragmented randomly by adding fragmentation buffer.
Blinding
not detected.
Power Analysis
not detected.
Replication
not required.
Cell Line Authentication
not detected.

Data Information
Availability: PR transgenic mouse model The PR transgenic mouse model, initially created by Shyamala et al,57 was resurrected from frozen embryos deposited at the Mutant Mouse Resource and Research Center (strain 032089-UCD) by the Lange and Hagan Labs.57 For immunophenotyping of mammary glands, inguinal lymph nodes, and spleens in 6-month-old transgenic mice and controls, mice were bred and aged for 6 months before they were sacrificed for tissue collection.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
Endogenous peroxidase was blocked with 3 H2O2 for 10 min followed by incubation with PR (1:500) primary antibody for 30 min (PR), followed by Mach 2 HRP Polymer (Biocare Medical) for 30 min and DAB +chromogen (Dako) for 5 min.	PR		
Experimental Models: Cell Lines			
E0771-vec and E0771-mPR cell lines were generated through lentiviral transduction using mouse progesterone receptor (mPR) or empty control vectors using pLenti-CMV-Hygro (Addgene 17454 , cloning details available on request) backbone.	E0771-mPR		
Experimental Models: Organisms/Strains			
70 71 E0771 tumor growth experiments P4 and placebo pellets (30mg, 30 day Innovative Research of America) were implanted into 8-week old C57BL6 mice obtained from the Jackson Laboratory (strain 000664).	C57BL6		Suggestion: RRID:IMSR_JAX:000664(link)
Recombinant DNA			
E0771-vec and E0771-mPR cell lines were generated through lentiviral transduction using mouse progesterone receptor (mPR) or empty control vectors using pLenti-CMV-Hygro (Addgene 17454 , cloning details available on request) backbone.	pLenti-CMV-Hygro		
Luciferase assays E0771 cell line variants were transiently transfected with a PRE-luciferase reporter construct (described in Faivre and Lange72) and pRL-TK , a Renilla luciferase construct used for normalizing transfection efficiency .	pRL-TK		
Software and Algorithms			
The tumor cells were then pelleted, supernatant was aspirated, and 2-5 mL of ACK Red Blood Cell Lysis Buffer (Gibco) was added to each sample.	Gibco		

Analysis: Gene Ontology (GO) and Gene Set Enrichment Analysis (GSEA) RNA sequences were aligned against mouse genome mm10 by STAR V.2.7.3 , and RNA expression was calculated by RNA-sequencing (RNA-Seq) by expectation maximization .	STAR		
The differentially expressed genes were found by using DESeq2 to compare RNA expression between controls and PR-overexpressing samples .	DESeq2		Suggestion: DESeq, RRID:SCR_000154(link)
Based on the statistics from the DESeq2 analysis , the significant HallMark pathways and GO pathways were found by GSEA.	GSEA		
The merged data were normalized by using the R limma package , and the batch effects between PAM50 and mouse data were removed by using SVA package.	limma		Suggestion: LIMMA (RRID:SCR_010943)(link)
Statistical software R or Prism V.6 (GraphPad Software) were used for all the analyses.	Prism		
	GraphPad		Suggestion: GraphPad Prism, RRID:SCR_002798(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Statistical analysis For tumor-free interval and overall survival comparison between the groups, Kaplan-Meier analyses followed by log-rank tests were carried out.	log-rank tests
Differences in the biomarkers between the two groups were assessed using two-sample t-tests or the Mann-Whitney test, depending on whether the data satisfied normality assumption or not.	t-tests
	Mann-Whitney test
Unresolved RRIDs	
E0771-vec and E0771-mPR cell lines were generated through lentiviral transduction using mouse progesterone receptor (mPR) or empty control vectors using pLenti-CMV-Hygro (Addgene 17454 , cloning details available on request) backbone.	Addgene 17454

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq